

CrabTransformer Training Report

Data: REAL | Scenario: REAL | Loss: MSE

— Real Data —

Source: Eastern Bering Sea Snow Crab Survey
Grid: 50x50 (25km cells, SPDE interpolation)
Valid ocean cells: 886
Bootstraps: 100 (subsampling from ~349 stations)

— Training Configuration —

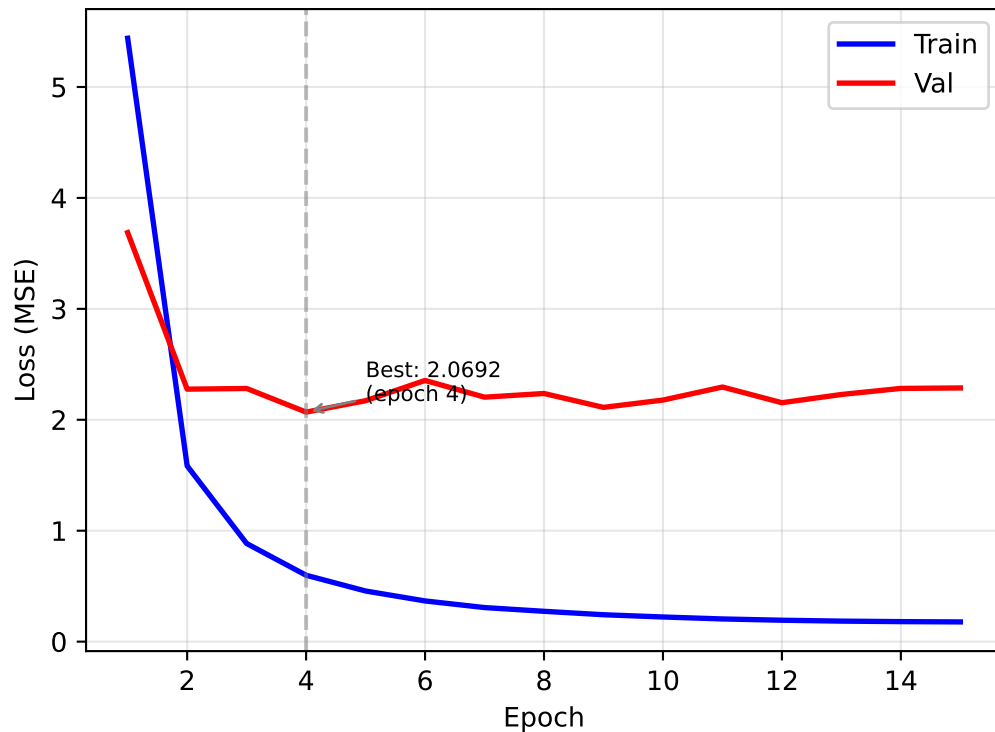
epochs_run: 15 best_val_epoch: 4
best_val_loss: 2.0692 final_train_loss: 0.1778
bias_correction (MSE): 1.2252

— Model Architecture —

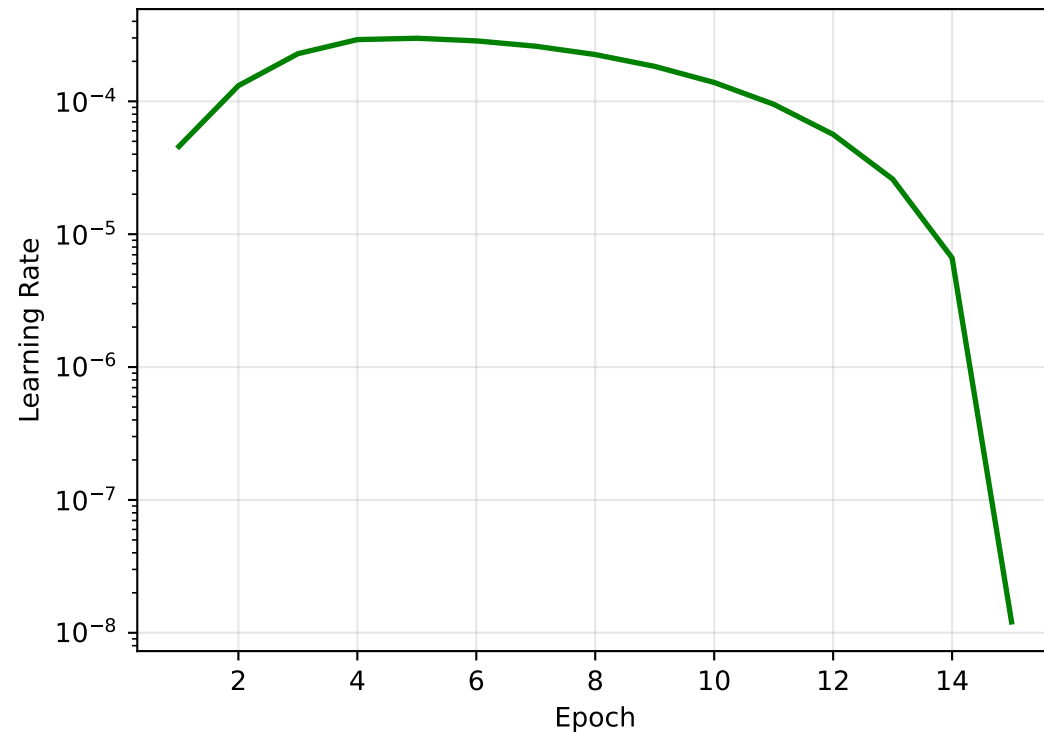
embed_dim: 128 num_heads: 8 num_layers: 6
d_ff: 512 dropout: 0.3 patch_size: 5
train/val/test years: 24/8/4

Training Curves: REAL | MSE

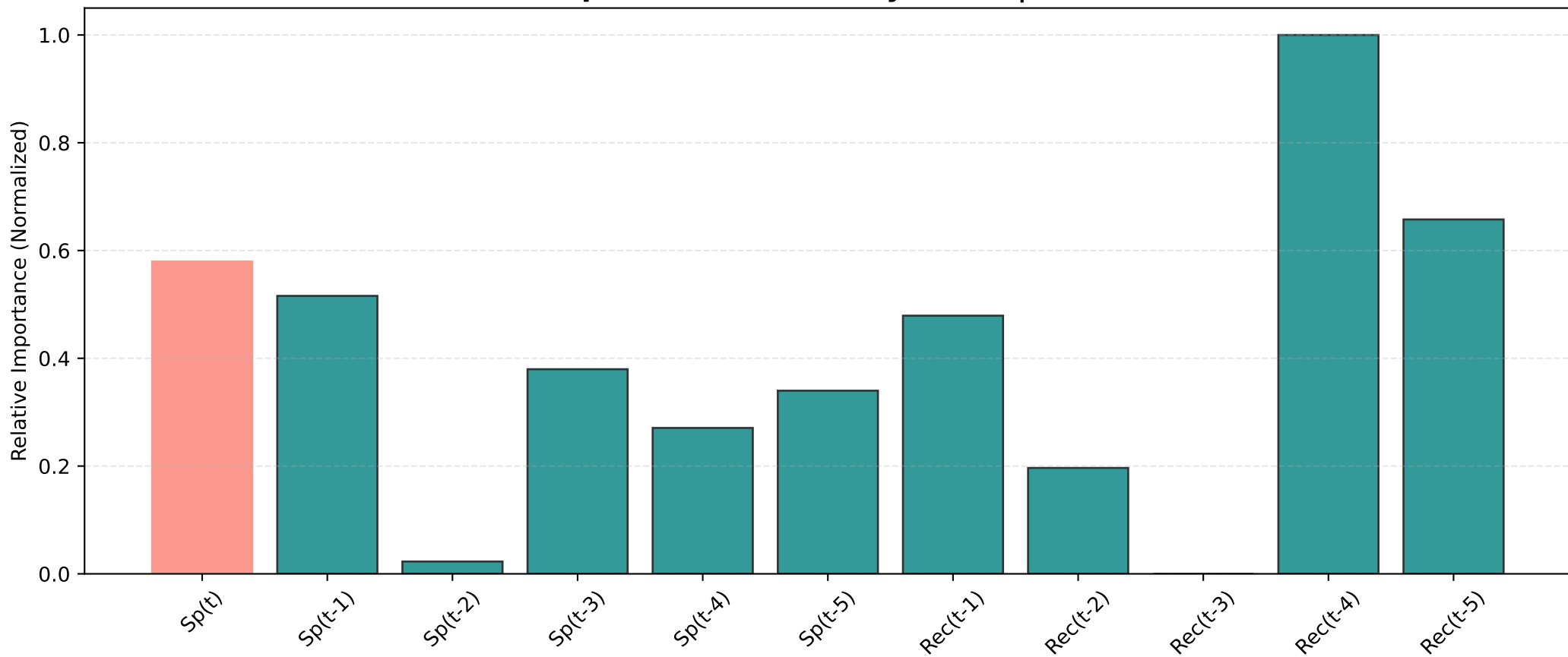
Training & Validation Loss



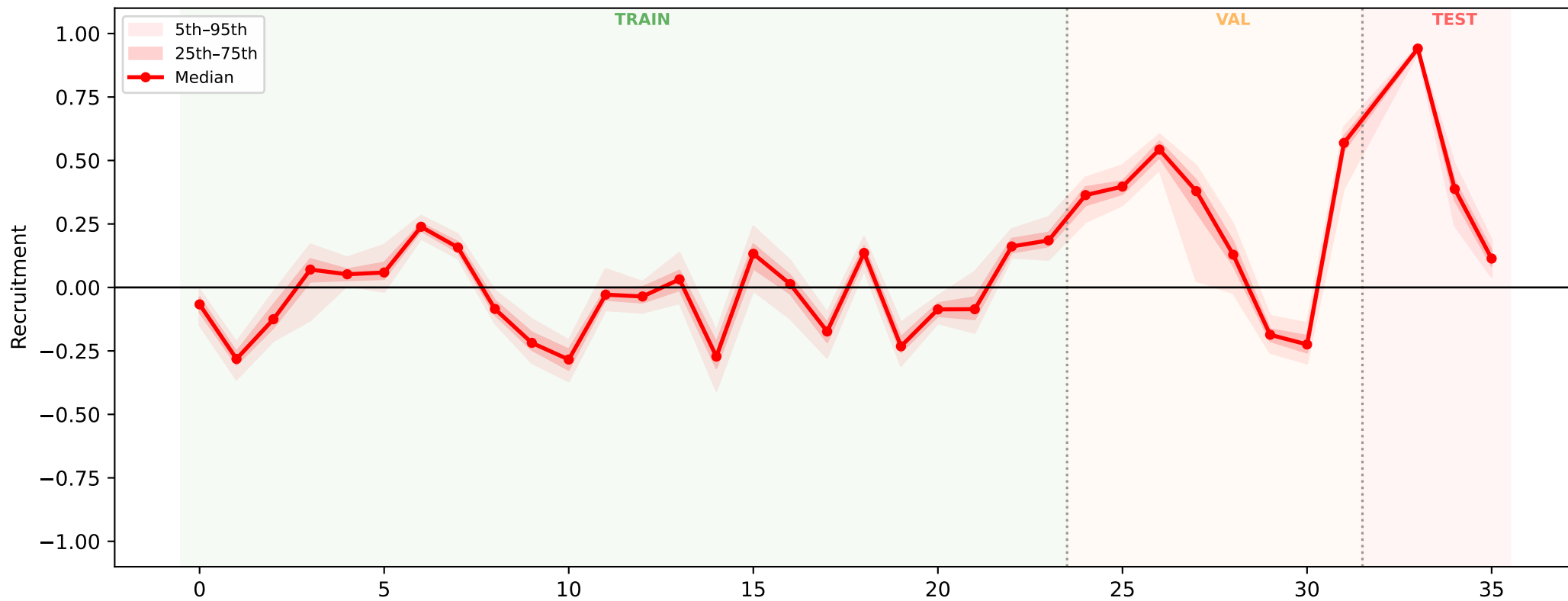
OneCycleLR Schedule



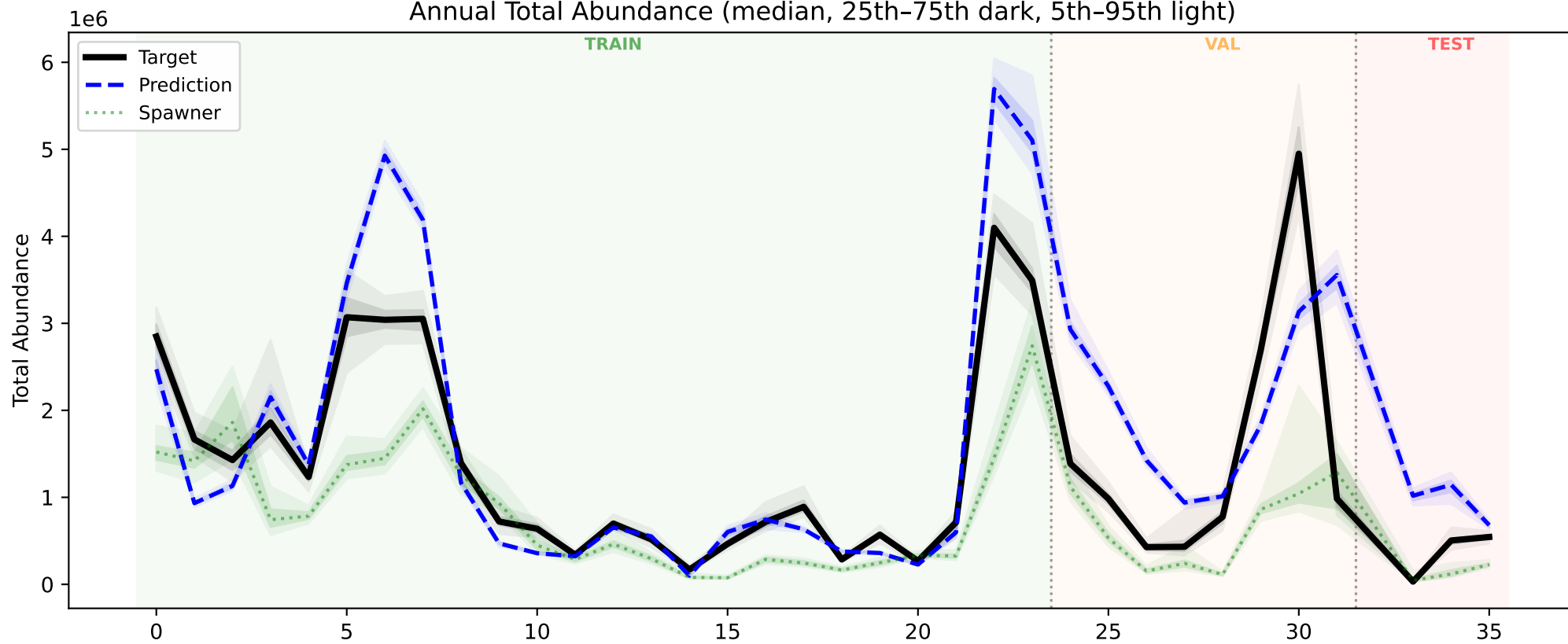
Input Channel Priority: REAL | MSE



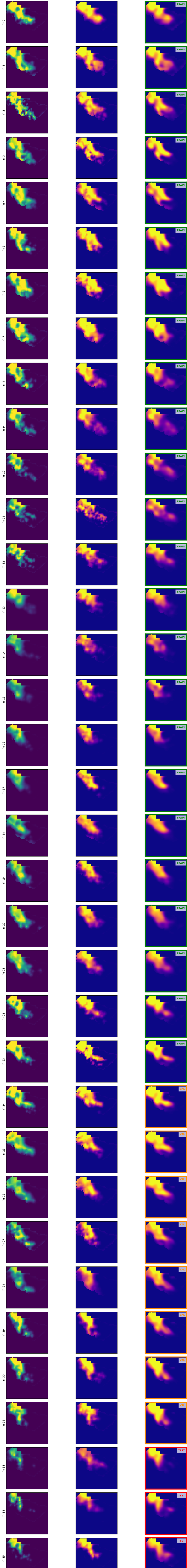
Annual Recruitment Deviations: REAL (MSE)



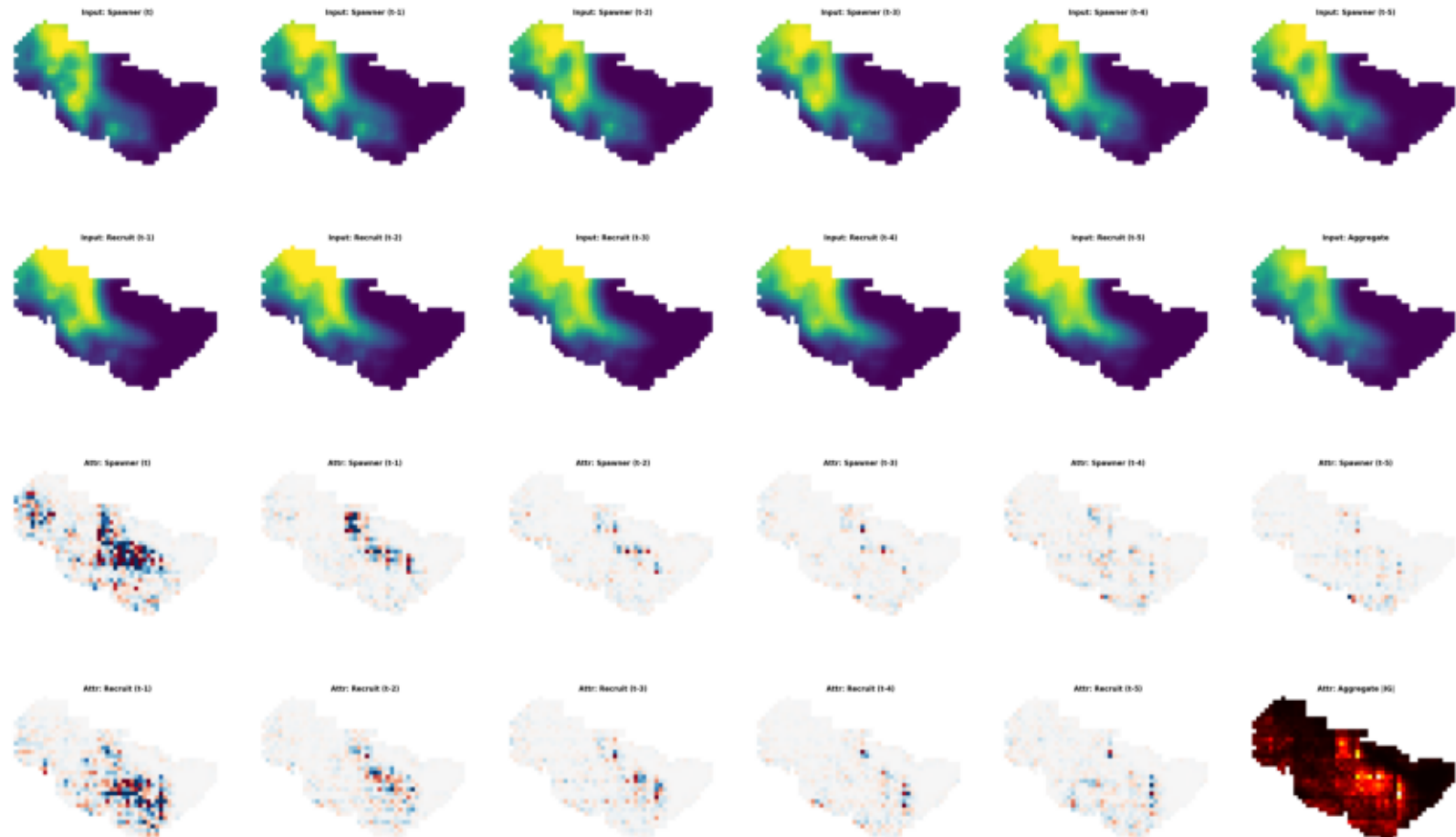
Annual Total Abundance (median, 25th-75th dark, 5th-95th light)



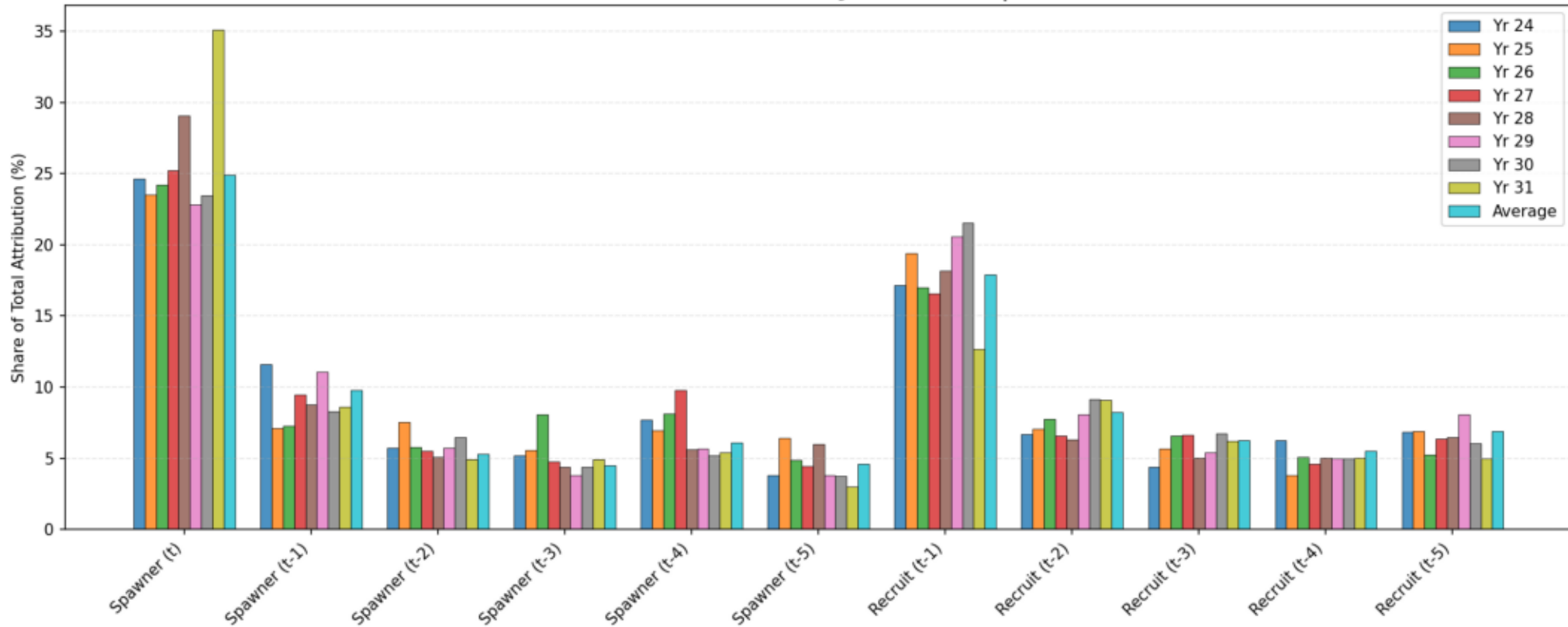
Spatial Trajectory: REAL | MSE



Year AVG — REAL | MSE
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Channel Attribution by Year: REAL | MSE



Year 24 — REAL | MSE
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 25 — REAL | MSE
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 26 — REAL | MSE
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 27 — REAL | MSE
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



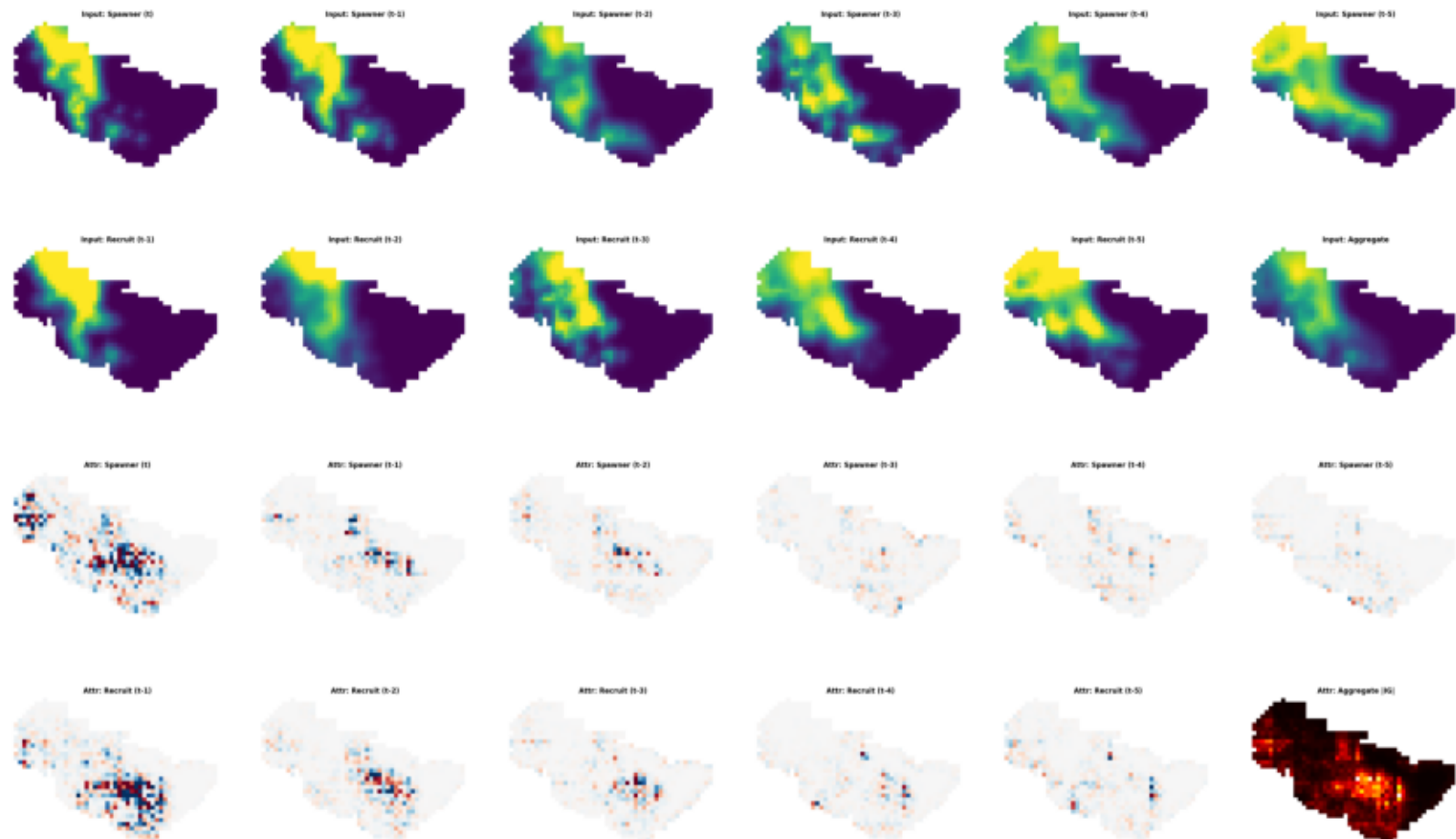
Year 28 — REAL | MSE
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



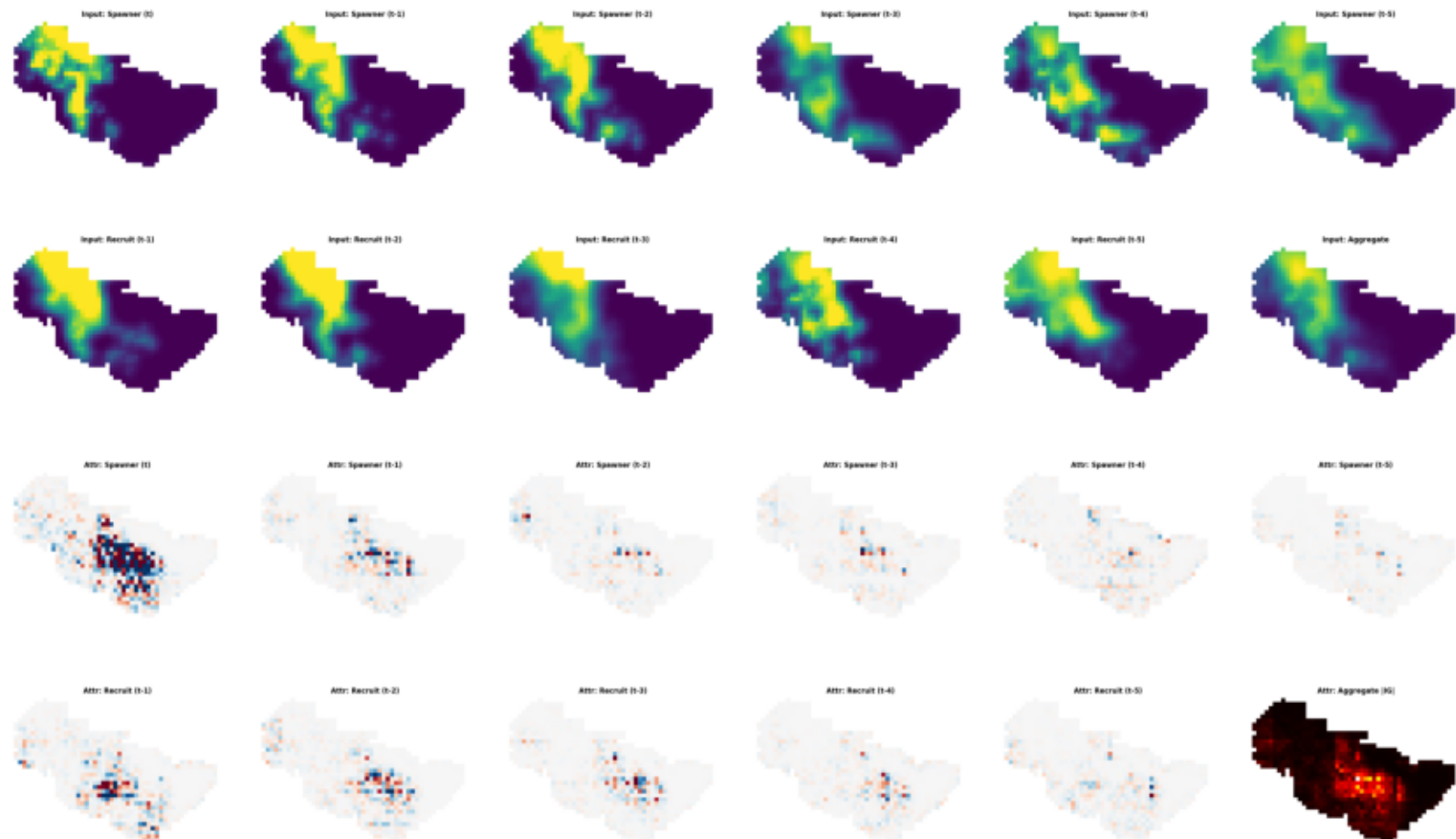
Year 29 — REAL | MSE
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 30 — REAL | MSE
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 31 — REAL | MSE
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Summary Statistics: REAL | MSE

Phase	Pred_Mean	Tgt_Mean	Pred_SD	Pred_P50	Pred_P95	Abundance	MAE	Spearman
TRAIN	1818.8	1612.8	7497.0	0.0	9641.2	1.128	859.2	0.956
VAL	2417.4	1801.5	7135.8	0.0	15881.9	1.342	2075.1	0.872
TEST	1058.7	406.3	3747.2	0.0	7021.6	2.606	744.5	0.868

Detailed Metrics (R-style): REAL | MSE

Phase	RMSE	RMSE_unbias	Bias	Bias_%	Spearman	Pearson	Pct_Error	Zero_F1	TopK_Acc	TopK_Jaccard	Quant_Err	Skill_MSE	Skill_MAE
TRAIN	2740.4	2686.1	206.0	3.1%	0.965	0.902	-1.8%	0.968	0.863	0.763	0.31	0.674	0.710
VAL	6320.6	6146.3	615.9	6.2%	0.876	0.674	96.6%	0.912	0.673	0.520	0.80	-2.107	0.029
TEST	2568.1	2479.6	652.3	5.6%	0.875	0.743	1133.1%	0.936	0.795	0.688	0.60	-571.122	-6.392

Appendix A: Summary Metrics

APPENDIX A: Table Metric Definitions

Data type: REAL
Split: 24 train / 8 val / 4 test years

Spatial masking: All metrics computed ONLY over valid ocean cells (~886) in the 50x50 grid. Padding cells are excluded from all computations.

All metrics computed on ORIGINAL DENSITY SCALE (after expm1 and bias correction for MSE). Zero threshold = 14 (smallest nonzero observed value): predictions AND observations below 14 are set to 0 before ALL metric computation. Both Page 7 and Page 8 apply this threshold.

Summary Table (Page 7)

Computed per bootstrap sample, then averaged across all samples in phase.

Pred_Mean, Tgt_Mean: Arithmetic mean of valid cell values.
Pred_SD: Standard deviation of valid cell values.
Pred_P50, Pred_P95: Median and 95th percentile.
Abundance: $\text{sum}(\text{pred}) / \text{sum}(\text{target})$. 1.0 = perfect.
MAE: $\text{mean}(|\text{pred} - \text{target}|)$ per valid cell, averaged across samples.
Spearman: Rank correlation over valid cells, averaged across samples.

Appendix A (cont): Detailed Metrics

APPENDIX A (cont): Detailed Metrics (Page 8)

Per-year: average spatial field across bootstraps, then compute metrics on that mean field over valid cells. Average across years per phase.

RMSE: $\sqrt{\text{mean}((\text{obs} - \text{pred})^2)}$ over valid cells.

RMSE_unbiased: RMSE after subtracting mean bias.

Bias: $\text{mean}(\text{pred} - \text{obs})$. Positive = overprediction.

Bias_%%: $\text{bias}^2 / \text{rmse}^2 * 100$.

Spearman / Pearson: Rank and linear correlation.

Pct_Error: $(\text{sum}(\text{pred}) - \text{sum}(\text{obs})) / \text{sum}(\text{obs}) * 100$.

Zero_F1: F1 for zero-classification (threshold = 14).

TopK_Accuracy / Jaccard: Overlap of top-10% density cells.

Quantile_Mean_Error: Mean |bin diff| across 10 quantile bins.

Skill_MSE / Skill_MAE: $1 - \text{error}/\text{climatology}$. Positive = beats mean.

Appendix B: Plot Methodology

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Page 2: Training Curves

MSE loss in log-space; Tweedie on original scale.

OneCycleLR: $\text{max_lr}/10 \rightarrow \text{max_lr} (10\%)$, cosine to $\text{max_lr}/1000$.

Page 3: Channel Importance

$\text{mean}(|\text{weights}|)$ of each channel's projection. Static, not causal.

Page 4: RecDev & Abundance

$\text{RecDev} = (\text{pred} - \text{target}) / (\text{target} + \text{pred})$. Sums over valid cells.

Ribbons: 5th-95th and 25th-75th percentiles across bootstraps.

Page 5: Spatial Trajectory

Log-space grids, first bootstrap, shared colorscale 0-8.

Page 6: Integrated Gradients

Pre-computed, averaged across bootstraps per year.

Output function sums over valid cells only (real data).

Bias Correction (MSE only)

$\text{pred} = \text{expm1}(\text{pred_log}) * \exp(\sigma^2 / 2)$

Tweedie: no correction needed.